Angles between lines

Basic skills

Q. a) $\begin{pmatrix}1\\4\end{pmatrix}$ b) $\begin{pmatrix}-1\\5\end{pmatrix}$

could have any scaled versions

c) $\begin{pmatrix}1\\5\\2\end{pmatrix}$ d) $\begin{pmatrix}-2\\-1\\0\end{pmatrix}$

e) $\underline{r} = \begin{pmatrix}1\\0\\0\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\3\end{pmatrix}$ so $\begin{pmatrix}1\\1\\3\end{pmatrix}$

f) $\underline{r} = \begin{pmatrix}0\\3\\2\end{pmatrix} + t\begin{pmatrix}2\\0\\-1\end{pmatrix}$ so $\begin{pmatrix}2\\0\\-1\end{pmatrix}$

Q2. a) $\underline{a} = \begin{pmatrix}1\\4\end{pmatrix}$ $\underline{b} = \begin{pmatrix}-1\\5\end{pmatrix}$

$$\cos\theta = \frac{\underline{a}\cdot\underline{b}}{|\underline{a}||\underline{b}|} = \frac{1\times -1 + 4\times 5}{\sqrt{1^2+4^2}\sqrt{1^2+5^2}} = 0.903$$

$$\theta = 25.35^\circ$$

could also haves $180 - 25.35$
$= 154.65$

b) $\underline{a} = \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$ $\underline{b} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$

$$\cos\theta = \frac{\underline{a}\cdot\underline{b}}{|\underline{a}||\underline{b}|} = \frac{1\times 2 + 5\times 1 + 2\times 0}{\sqrt{1^2+5^2+2^2}\sqrt{2^2+1^2}} = \frac{7}{3\sqrt{30}}$$

$$\theta = 64.8^\circ$$

Competency skills

Q3. $\underline{r}_1 = \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix} + \lambda\begin{pmatrix} 6 \\ -1 \\ 1 \end{pmatrix}$ $\underline{r}_2 = \begin{pmatrix} 7 \\ 7 \\ 1 \end{pmatrix} + \mu\begin{pmatrix} 5 \\ 0 \\ 1 \end{pmatrix}$

a) $\underline{d}_1 = \begin{pmatrix} 6 \\ -1 \\ 1 \end{pmatrix}$ $\underline{d}_2 = \begin{pmatrix} 5 \\ 0 \\ 1 \end{pmatrix}$

b) $$\cos\theta = \frac{\underline{d}_1\cdot\underline{d}_2}{|\underline{d}_1||\underline{d}_2|} = \frac{30+1}{\sqrt{6^2+1^2+1^2}\sqrt{5^2+1^2}} = \frac{31}{2\sqrt{247}}$$

$$\theta = 9.52^\circ$$

c) For obtuse $180 - 9.52^\circ$

$= 170.48^\circ$

Q4. a) $\underline{d}_1 = \begin{pmatrix} 5 \\ 1 \\ 2 \end{pmatrix}$ $\underline{d}_2 = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$

$$\cos\theta = \frac{\underline{d}_1 \cdot \underline{d}_2}{|\underline{d}_1||\underline{d}_2|} = \frac{-5+2+2}{\sqrt{5^2+1^2+2^2}\sqrt{1^2+2^2+1^2}} = -\frac{1}{6\sqrt{5}}$$

$\theta = 94.3^\circ$

For acute $180 - 94.3^\circ = 85.7^\circ$

b) $\underline{d}_1 = \begin{pmatrix} 5 \\ 10 \\ -2 \end{pmatrix}$ $\underline{d}_2 = \begin{pmatrix} -5 \\ -10 \\ 2 \end{pmatrix}$

As $\underline{d}_1 = -\underline{d}_2$ lines are parallel so angle is 0°

Normal method should give $\cos\theta = \pm 1$

c) $\underline{d}_1 = \begin{pmatrix} 2 \\ 2 \\ -4 \end{pmatrix}$ $\underline{d}_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$

$$\cos\theta = \frac{2+4-4}{\sqrt{24}\sqrt{6}} = \frac{1}{6} \qquad \theta = 80.4^\circ$$

d) $\underline{d}_1 = \begin{pmatrix} 2 \\ -2 \\ 4 \end{pmatrix}$ $\underline{d}_2 = \begin{pmatrix} 2 \\ -3 \\ 2 \end{pmatrix}$ ← denominators

$$\cos\theta = \frac{4+6+8}{\sqrt{24}\sqrt{17}} = \frac{3\sqrt{102}}{34} \approx 0.8911...$$

$$\theta = 26.98...$$
$$= 27.0^\circ$$

Advanced skills

Q5. $\underline{d}_1 = \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix}$ $\underline{d}_2 = \begin{pmatrix} k \\ 1 \\ 1 \end{pmatrix}$ $\theta = 60^\circ$

$$\cos 60^\circ = \frac{2k+5}{\sqrt{29}\sqrt{k^2+2}}$$

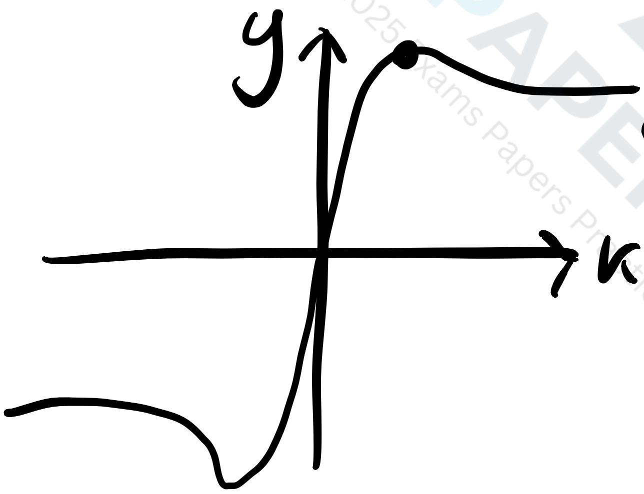
SO $\frac{1}{2} = \frac{2k+5}{\sqrt{29k^2+58}} \Rightarrow \sqrt{29k^2+58} = 4k+10$

$$\Rightarrow 29k^2+58 = (4k+10)^2$$
$$\Rightarrow 29k^2+58 = 16k^2+80k+100$$
$$\Rightarrow 13k^2-80k-42=0$$
$$\Rightarrow k = \frac{80 \pm \sqrt{80^2+4\times13\times42}}{2\times13} = 6.64 \text{ or } -0.49$$

Q6. $\underline{d}_1 = \begin{pmatrix} 3k \\ 2 \\ 1 \end{pmatrix}$ $\underline{d}_2 = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$

$$\cos\theta = \frac{3k - 2 + 3}{\sqrt{9k^2+5}\sqrt{11}} = \frac{3k+1}{\sqrt{99k^2+55}}$$

For minimal angle want as close to 1 as possible.



$y = \dfrac{3k+1}{\sqrt{99k^2+55}}$ want max to get as close to 1.

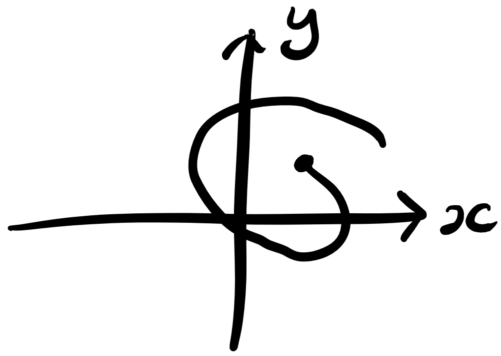
$$y' = \frac{15 - 9k}{(99x^2+55)^{\frac{3}{2}}}$$ so $15 - 9k = 0$

$k = \frac{5}{3}$

giving $y = \dfrac{5+1}{\sqrt{99(\frac{5}{3})^2+55}} = \sqrt{\dfrac{6}{55}} < 1.$

so $k = \frac{5}{3}$

Q 7. $\underline{r} = e^{t}\begin{pmatrix} 1+\sin t \\ \cos t \end{pmatrix}$



a) x-axis:
$y=0$ when

$\cos t = 0 \quad t = \frac{\pi}{2} + \pi k$

y-axis:
$x=0$ when

$1+\sin t = 0 \quad \sin t = -1, \quad t = -\frac{\pi}{2} + 2\pi k$

b) First ints at $t = \frac{\pi}{2}$, $t = \frac{3\pi}{2}$

Points $(e^{\frac{\pi}{2}}, 0)$, $(0, 0)$

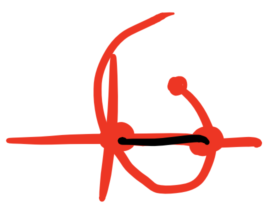
so $\underline{r} = \lambda\begin{pmatrix} e^{\frac{\pi}{2}} \\ 0 \end{pmatrix}$

c) x-axis is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ y-axis $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\cos\theta = \frac{e^{\frac{\pi}{2}}}{\sqrt{(e^{\frac{\pi}{2}})^2}} = 1$

$\cos\theta = 0$

$\theta = 90^\circ$



$\theta = 0^\circ$ ← parallel to x-axis